SIMATS ENGINEERING

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

LAB EXPERIMENTS

CSA1514 CLOUD COMPUTING

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BRANCH: B. TECH AIML

Experiment 1: Flight Reservation System

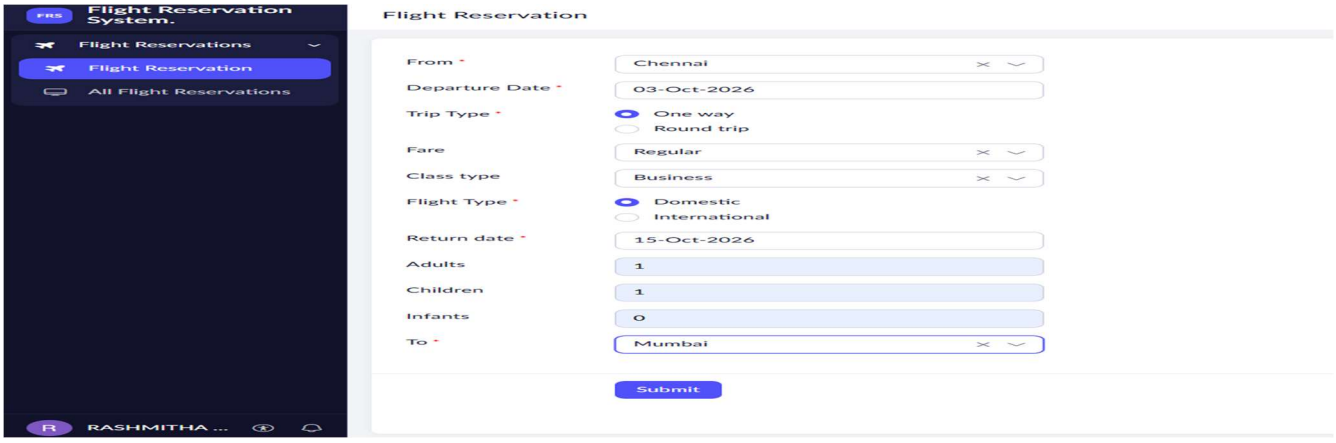
Aim:

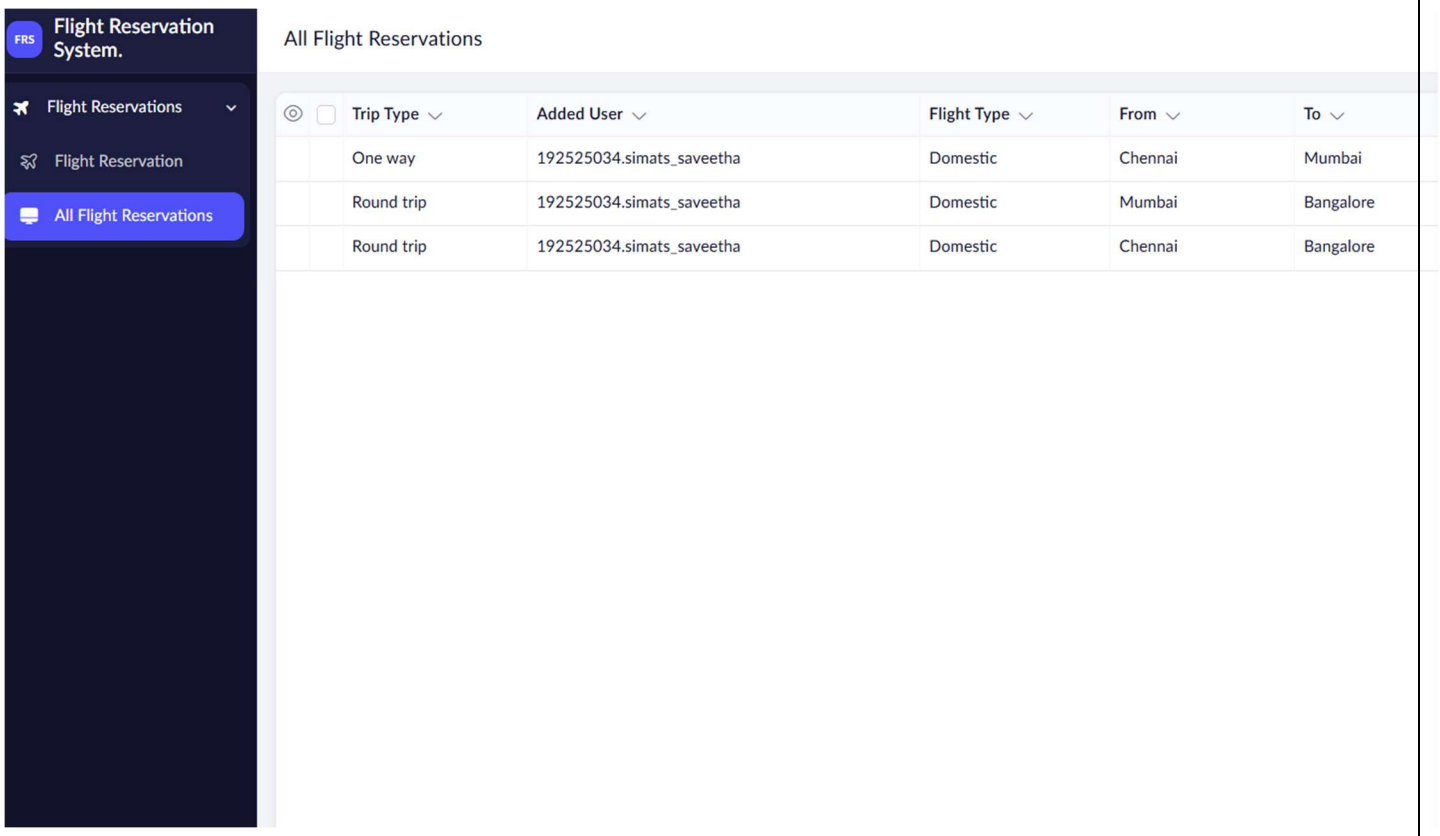
To create a cloud-based application for storing flight reservation details using Zoho Creator.

Procedure:

1. Create a Zoho Creator application named Flight Reservation System and a Flight Reservation form.
2. Add Trip Type and Flight Type as radio fields; From, To, Class and Fare Type as dropdown fields.
3. Add Departure Date and Return Date as date fields, and Adults, Children and Infants as number fields. Keep Return Date optional for one-way travel.
4. Enter a sample one-way domestic booking from Chennai to Delhi for one adult in Economy class.
5. Click Submit and open the report to verify the saved booking.

Output:





Result:

After successful execution: The flight reservation application stores and displays booking details.

Experiment 2: Property Buying and Rental System

Aim:

To create a cloud-based application for recording property buying and rental details using Zoho Creator.

Procedure:

1. Create an application named Chennai Property Buying and Rental System and a Property Details form.

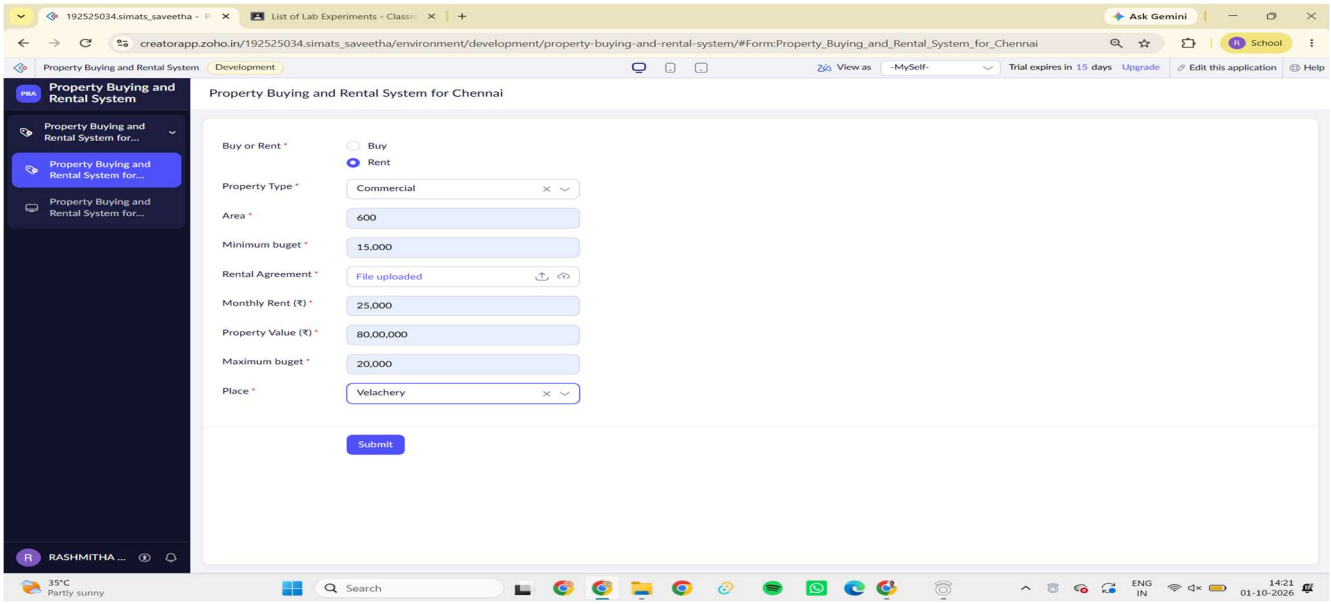
2. Add Buy or Rent as a radio field, Property Type and Place as dropdowns, and Area in Square Feet as a number field.

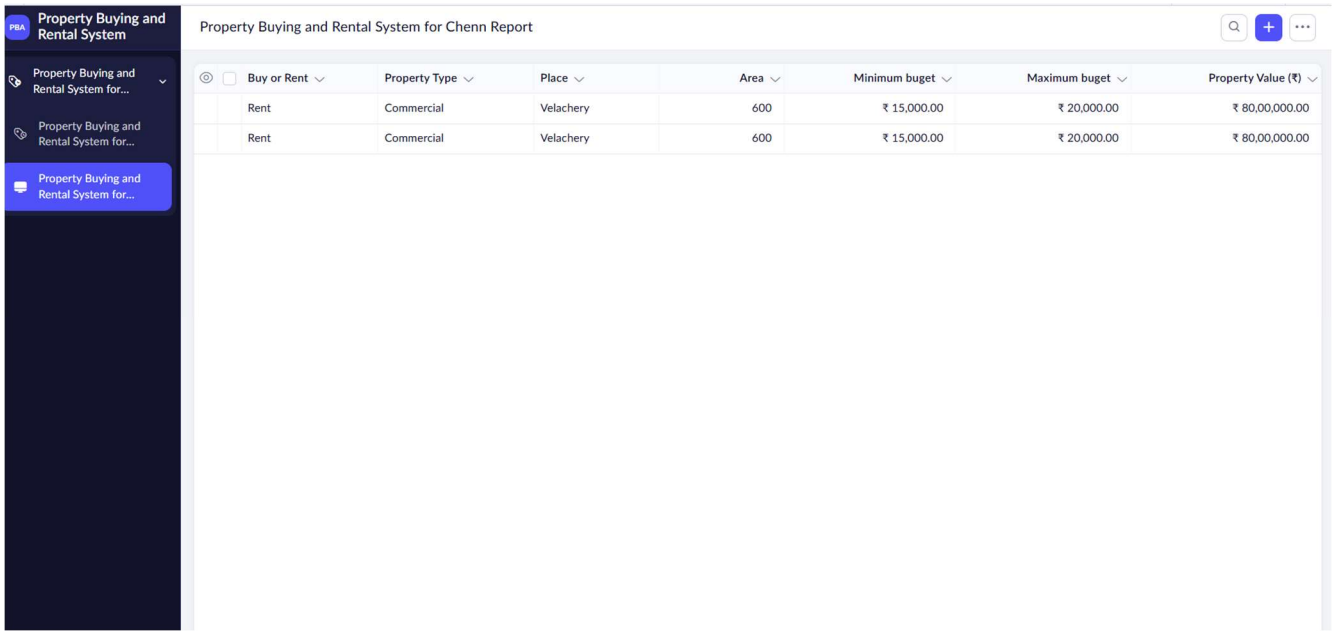
3. Add Minimum Budget, Maximum Budget, Property Value and Monthly Rent as currency fields; add Rental Agreement as a file upload field.

4. Enter a sample residential rental in Anna Nagar with an area of 1,000 square feet and monthly rent of Rs. 20,000. Upload a sample agreement if applicable.

5. Submit the form and verify the saved record in the report.

Output:





Result:

After successful execution: The application stores and displays property buying or rental details.

Experiment 3: Car Booking Reservation System

Aim:

To create a cloud-based car reservation application using Zoho Creator.

Procedure:

1. Create a Car Booking Reservation System application and a Car Booking form.

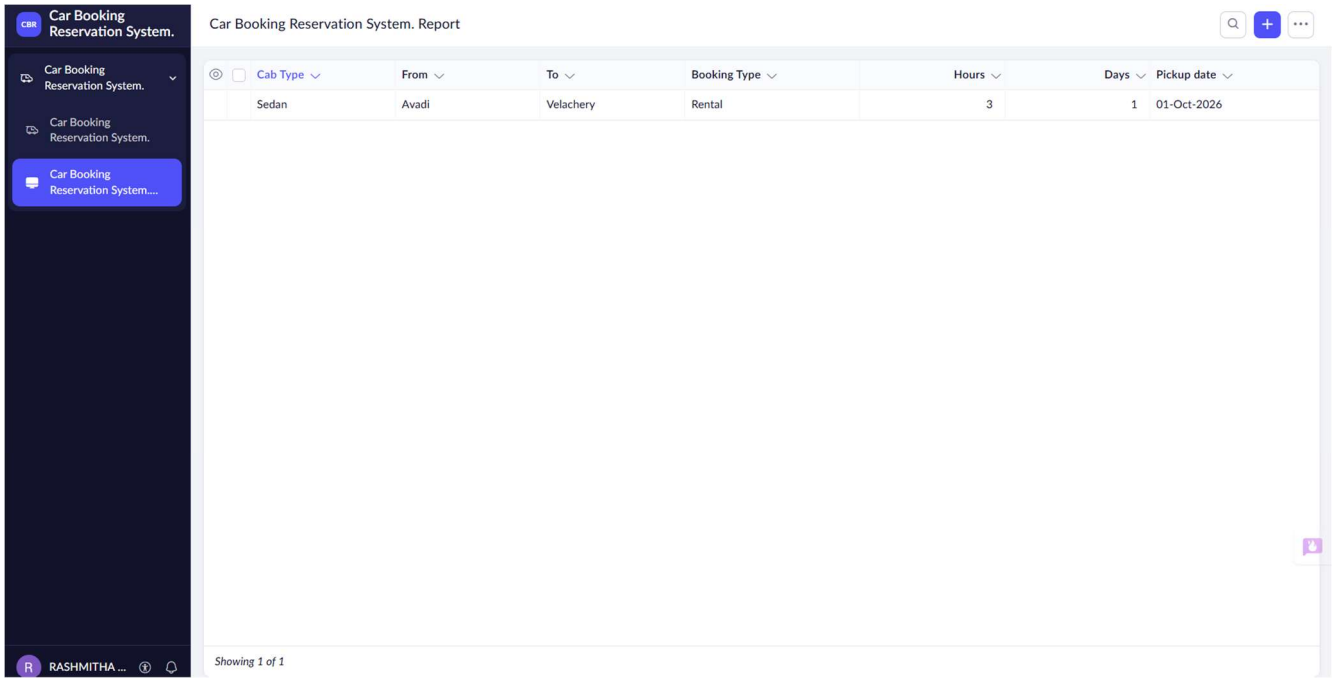
2. Add Cab Type and Package Type as dropdowns, From and To as single-line fields, and Booking Type as a radio field.

3. Add Hours or Days as a number field, Pickup Date as a date field and Pickup Time as a time field.

4. Enter a Sedan booking from Chennai Airport to Anna Nagar, with Rental, Hourly and four hours selected. Enter a pickup date and time.

5. Submit the booking and verify its details in the report.

Output:



Result:

After successful execution: The car reservation application stores and displays booking details.

Experiment 4: Library Book Reservation System

Aim:

To create a cloud-based application for recording library book details and borrowing status.

Procedure:

1. Create a SIMATS Library Book Reservation System application and a Book Reservation form.

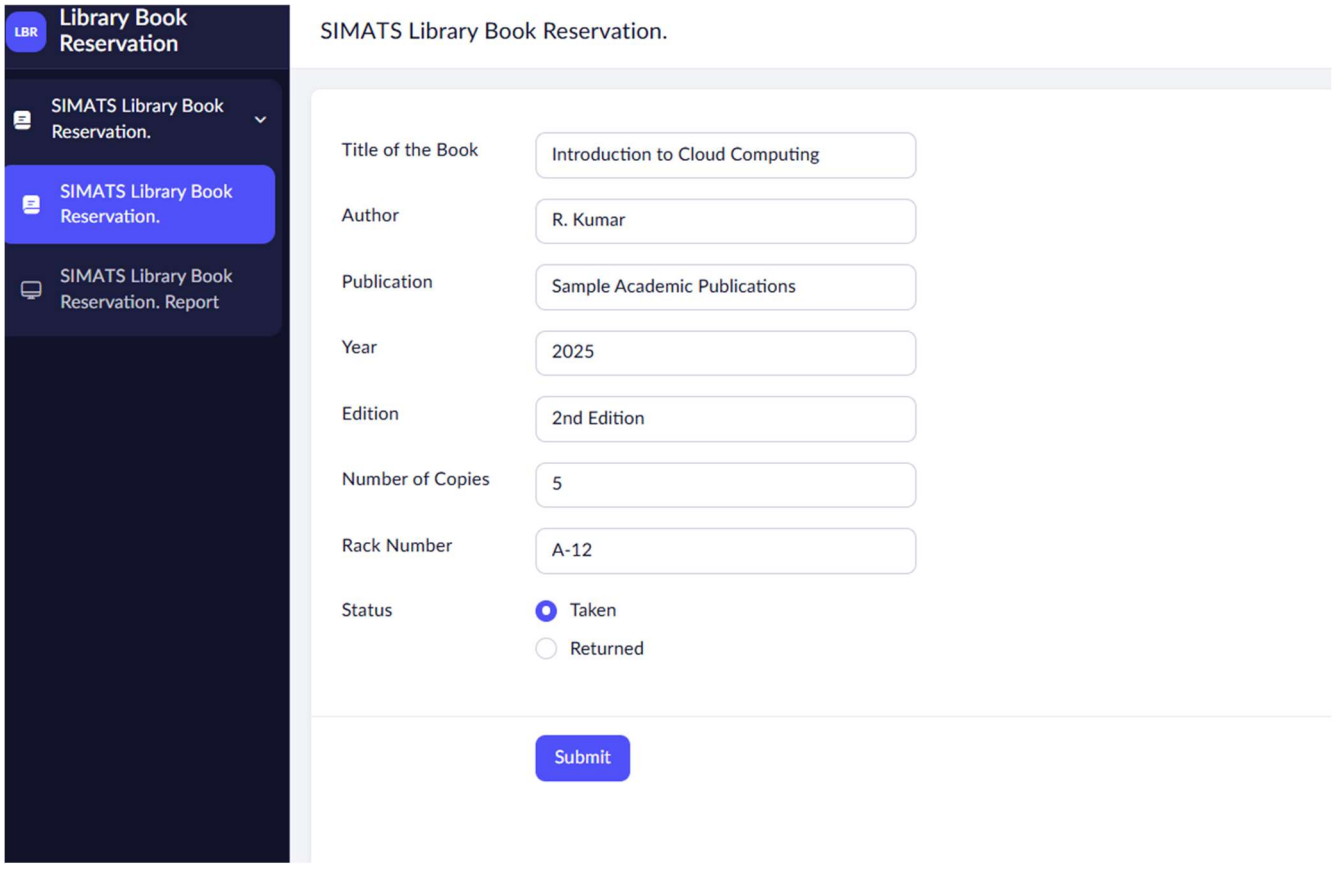
2. Add Book Title, Author, Publication, Edition and Rack Number as single-line fields.

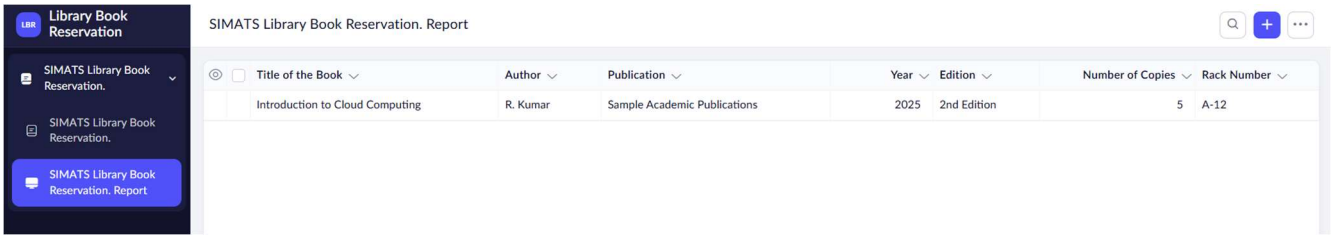
3. Add Year of Publication and Number of Copies as number fields; add Status as a radio field with Taken and Returned.

4. Enter fictional sample book details, select Taken and submit the record.

5. Open the report, edit the record to Returned when applicable, and verify the updated status.

Output:





Result:

After successful execution: The library application records book details and updates borrowing status.

Experiment 5: Mark Sheet Processing System

Aim:

To create a cloud-based mark sheet application that calculates total marks, average and percentage.

Procedure:

1. Create a Mark Sheet Processing System application and a Student Mark Sheet form.

2. Add Student Name and Register Number as single-line fields; add Mathematics, Computer Science and Cloud Computing as number fields.

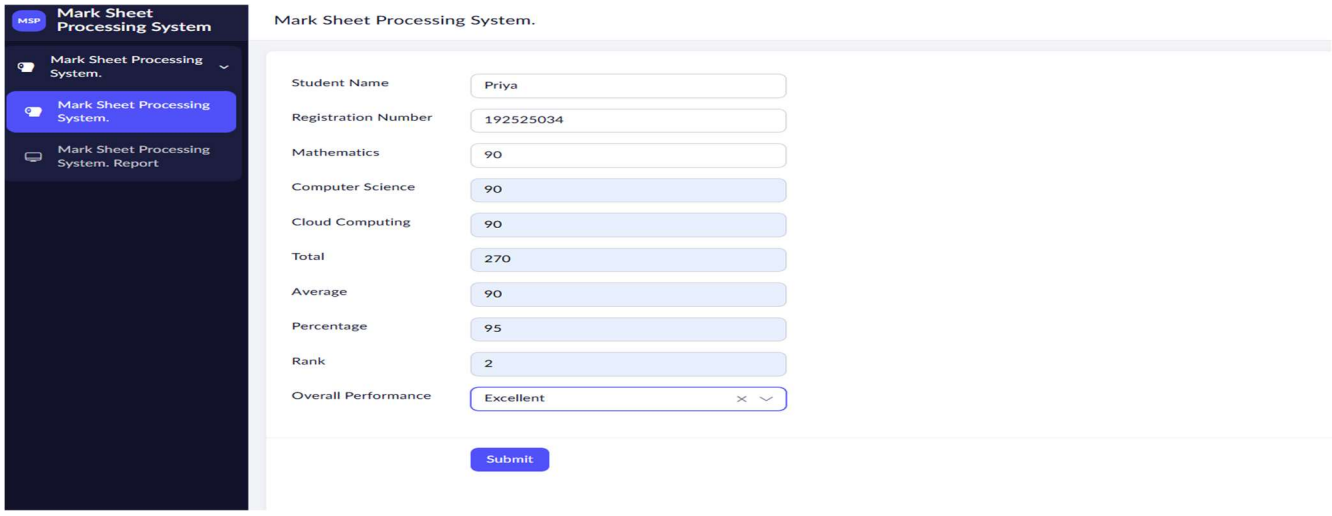
3. Add Total Marks, Average and Percentage as formula fields. Use the exact field link names from the formula editor.

4. Calculate Total Marks as the sum of the three marks; Average as the sum divided by 3.0; Percentage as the sum multiplied by 100.0 and divided by 300.

5. Add Rank as a number field and Overall Performance as a dropdown. Enter rank only after comparing student results; use institutional performance rules.

6. Enter marks of 80, 90 and 85, submit the form and check the calculations. Each subject is out of 100.

Output:



Result:

After successful execution: The application calculates and stores total marks, average and percentage.

Experiment 6: Payroll Processing System

Aim:

To create a cloud-based payroll application that calculates gross pay and net pay.

Procedure:

1. Create a Payroll Processing System application and an Employee Payroll form.

2. Add Employee Name as a single-line field and Present Organisation Experience and Overall Experience as decimal fields.

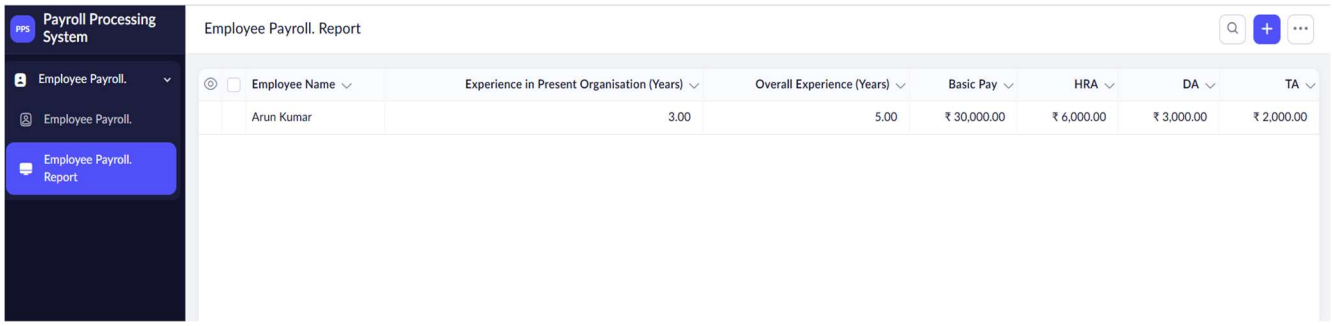
3. Add Basic Pay, HRA, DA, TA and Deductions as currency fields; add Gross Pay and Net Pay as formula fields.

4. Set Gross Pay = Basic Pay + HRA + DA + TA. Set Net Pay = Basic Pay + HRA + DA + TA - Deductions, using the actual field link names.

5. Enter Basic Pay 30000, HRA 6000, DA 3000, TA 2000 and Deductions 1000. These are sample amounts.

6. Submit the form and check the calculated salary in the report.

Output:



Result:

After successful execution: The payroll application calculates gross and net pay from the entered amounts.

Experiment 7: Student Attendance System

Aim:

To create a cloud-based student attendance application that calculates attendance percentage.

Procedure:

1. Create a Student Attendance System application and a Student Attendance form.

2. Add Student Name and Grade as single-line fields, Course as a dropdown, and Attendance Date as a date field.

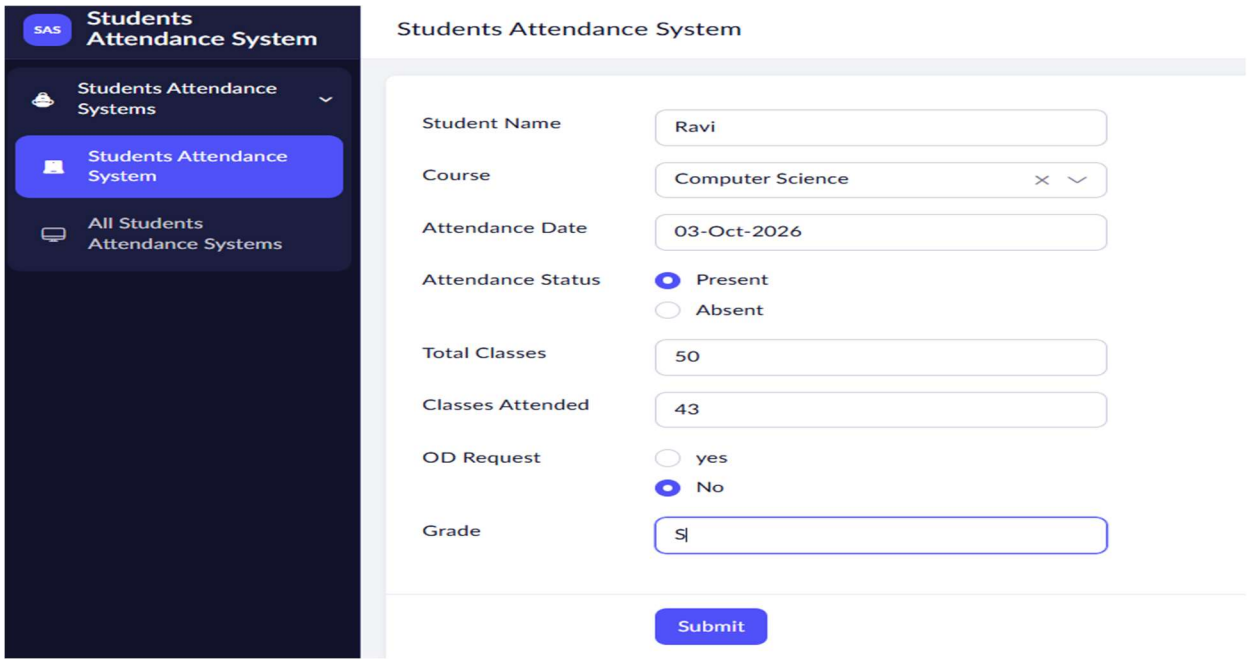
3. Add Attendance Status as a Present or Absent radio field, OD Request as a Yes or No radio field, and Total Classes and Classes Attended as number fields.

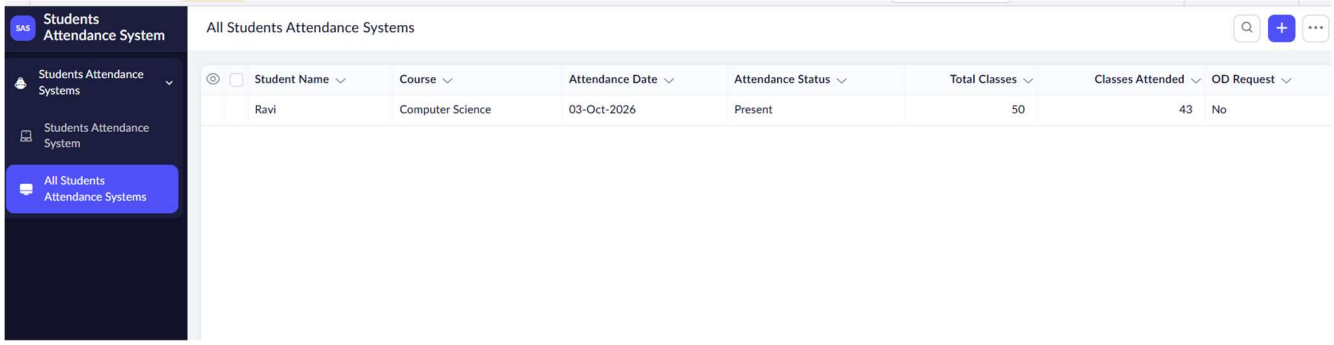
4. Add an Attendance Percentage formula field. Calculate Classes Attended * 100.0 / Total Classes using the correct field link names.

5. Ensure Total Classes is greater than zero and Classes Attended is between zero and Total Classes. Enter 30 total classes and 24 attended.

6. Select the remaining details, submit and check the report. Enter grade only according to institutional rules.

Output:





Result:

After successful execution: The application stores attendance details and calculates attendance percentage.

Experiment 8: Student Counselling System

Aim:

To create a cloud-based application for collecting student details and academic interests.

Procedure:

1. Create a Student Counselling System application and a Student Details form.

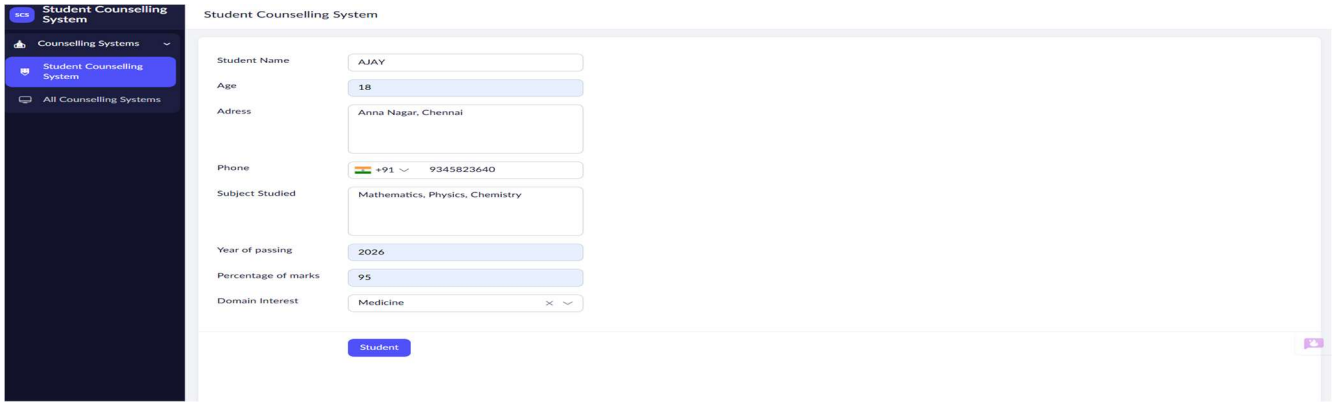
2. Add Student Name as a single-line field, Age and Year of Passing as number fields, and Address and Subjects Studied as multi-line fields.

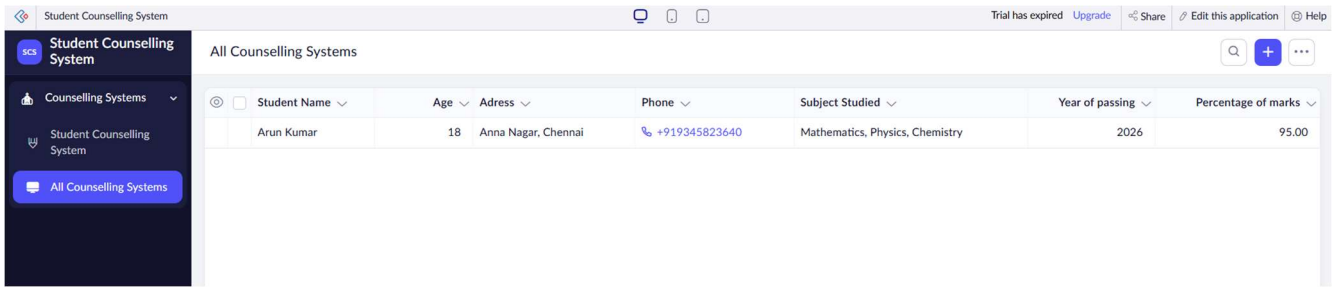
3. Add Phone as a phone field, Percentage of Marks as a decimal field and Domain Interest as a dropdown with Engineering, Medicine, Arts and Commerce.

4. Enter fictional sample details: Arun Kumar, age 18, Chennai, Mathematics, Physics and Chemistry, passing year 2026, marks 85.5% and Engineering interest.

5. Enter a test phone number, submit the form and view the saved record in the report.

Output:





Result

After successful execution: The counselling application stores and displays student details and interests.

Experiment 9: Hospital Information System

Aim:

To create a cloud-based hospital application for storing patient, attender and doctor details.

Procedure:

1. Create a SIMATS Hospital Information System application and a Patient Details form.

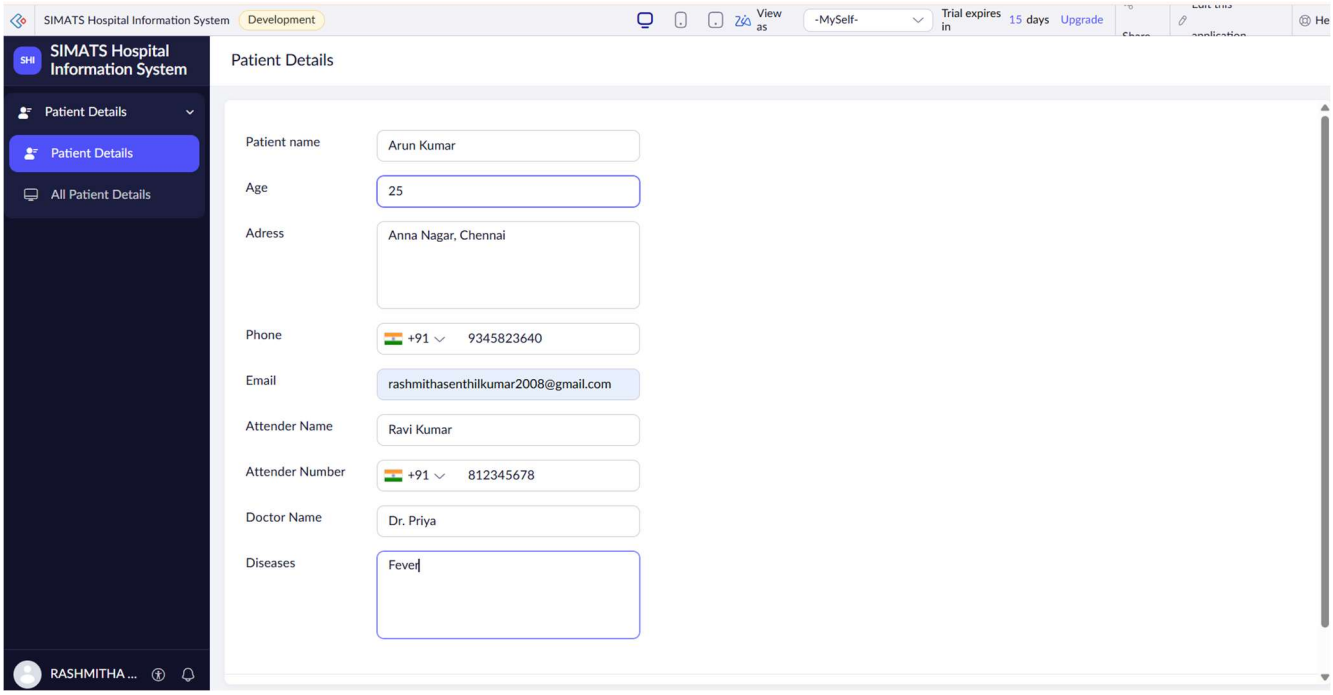
2. Add Patient Name, Attender Name and Doctor Name as single-line fields, and Age as a number field.

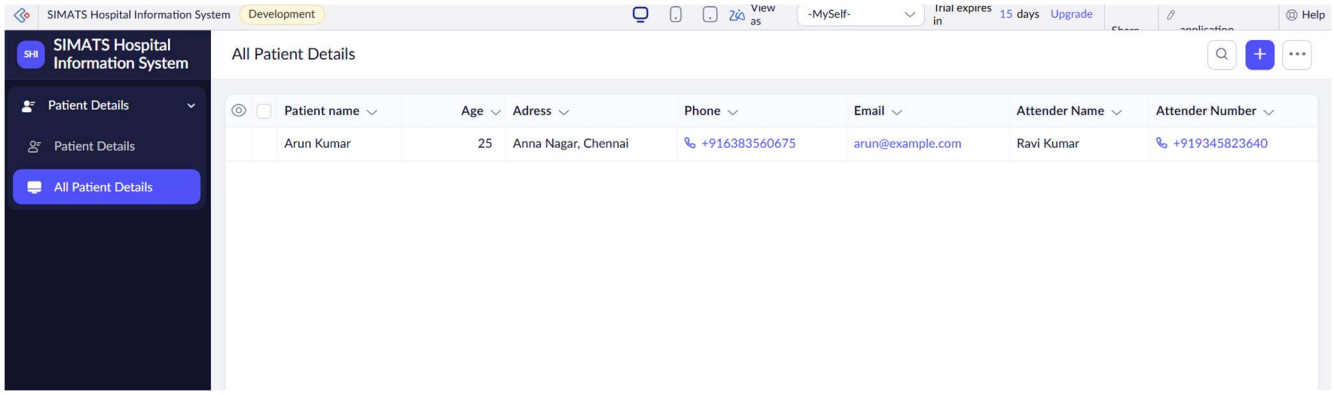
3. Add Address and Diseases as multi-line fields, Phone and Attender Phone as phone fields, and Email as an email field.

4. Enter fictional test details: Arun Kumar, age 25, Chennai, arun@example.com, attender Ravi Kumar, doctor Dr. Priya and disease Fever. Use test phone numbers.

5. Submit the form and verify all entered details in the patient report.

Output:





Result:

After successful execution: The hospital application stores and displays patient-related information.

Experiment 10: Student Information Using a Cloud C Compiler

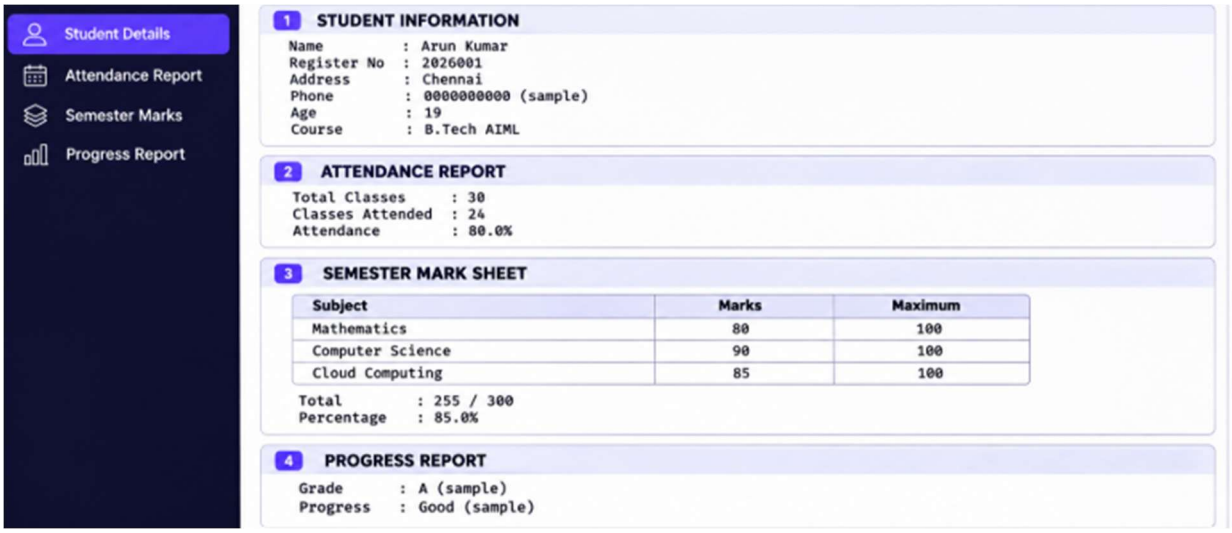
Aim:

To develop and execute a C program using an online cloud compiler to display student information, attendance percentage, semester marks and progress details.

Procedure:

1. Open OnlineGDB and select C as the programming language.
2. Enter the C program containing the student's name, register number, address, phone number, age and course.
3. Include total classes and classes attended to calculate attendance percentage.
4. Enter three subject marks and calculate their total and percentage.
5. Include a sample grade and progress remark.
6. Click Run to compile and execute the program.
7. Verify the displayed information and capture the output screenshot.

Output:



Result:

On successful execution, the program displays student details, attendance and academic progress. For the sample values, attendance is 80%, total marks are 255 out of 300, and marks percentage is 85%.

Experiment 11 - Online Supermarket System

Aim:

To create a cloud-based supermarket application for recording customer orders and classifying items.

Procedure:

1. Create an Online Supermarket application and a Customer Order form.

2. Add Customer Name as a single-line field, Address as multi-line, Phone as phone, Email as email, Item as a dropdown and Quantity as a number field.

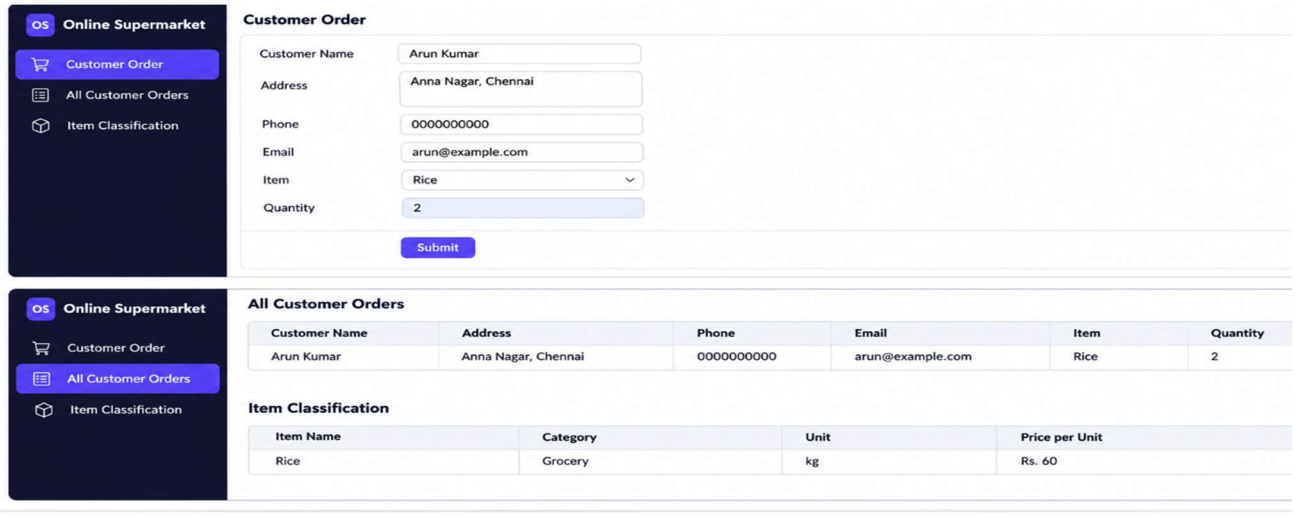
3. Create an Item Classification form with Item Name as single-line, Category and Unit as dropdowns, and Price per Unit as currency.

4. Use categories Grocery, Dairy, Personal Care and Fruits, and units kg, litre and piece.

5. Save a sample classification: Rice, Grocery, kg, Rs. 60 per unit. Save a customer order for Arun Kumar, Chennai, arun@example.com, Rice and quantity 2 using a test phone number.

6. Open both reports and verify the records. These forms record orders and items; payment and delivery are outside this demonstration.

Output:



Result:

After successful execution: The supermarket application stores customer orders and item classifications.